

1 Class Questions

$\in$: ‘is a’ or ‘is an element of’

$$a \in \{a, b, c\}$$

$\{a\} \in \{a, b, c\}$? There are many canine pets, let’s call the collection of them dogs. My pets name is Maeby, she is a dog, that is, Maeby is an element of the set of dogs. Is the collection that only includes Maeby a dog?

1.1 Breakout Sessions

$$a \in \{\{a, b\}, c\}$$

$$c \in \{a, b, \{c\}, c\}$$

$$\{a, b\} \in \{\{a, b\}, c\}$$

$$\{b\} \in \{\{a, b\}, c\}$$

$$[4, 3) \in \mathbf{R}$$

Sketching Functions

$$f(x) = x^3 - 2x + 8$$

$$f(x) = 3x^2 - \log(x) + 2$$

$$f(x) = 1 - 2 * \log(x/2)$$

$$f(x) = \frac{1}{3x^2} + 2$$

2 Set Equality

We introduced in lecture the concept of a set, and the constructor notation
e.g. $A = \{x : x \in [0, 1]\}$

The following sounds pedantic:

- A set A is = to a set B if they have exactly the same elements, $x \in A$ iff $x \in B$.
- A set A is subset of B if every element of A is in B .

But from here it follows that if A set A is subset of B and B is subset of A , then $A = B$ and vice versa.

How do we know? Start from the left, use the definitions: Pick an arbitrary element of A , by the definition of subset, it is in B . So $x \in A \rightarrow x \in B$. Pick an arbitrary element of B , by the definition of subset, it is in A . So $x \in B \rightarrow x \in A$. We are all set.

2.1 Example 1: Distributive property

A non-empty collection of sets is called a "class" or "family"; where $A \in \mathcal{A}$ are sets.

The union of all member of a class is denoted $\cup \mathcal{A}$

$$\cup \{A : A \in \mathcal{A}\}$$

Let A be a set and $\mathcal{B}$ be a class of sets: Prove that

$$A \cap \cup \mathcal{B} = \cup \{A \cap B : B \in \mathcal{B}\}$$

$$Z = Q$$

We are asked to prove that two sets are equal, they are if the first is a subset the second and the second is a subset of the first.

Part 1 $A \cap \cup \mathcal{B} \subseteq \cup \{A \cap B : B \in \mathcal{B}\}$

Pick $x \in Z$, we want to show that it is in Q

$x \in Z$, so $x \in A$ and $x \in B_i$ call that B_1

$$(A \cap B_1)$$

so an element of a set is also the element of that set union some other set:

$$x \in (A \cap B_1) \cup (A \cap B_2) \dots$$

$$\cup \{A \cap B : B \in \mathcal{B}\}$$

Part 2: $A \cap \cup \mathcal{B} \supseteq \cup \{A \cap B : B \in \mathcal{B}\}$

Pick $x \in Q$, we want to show that it is in Z . To do that we need to show that x is in A and x is in $\cup \mathcal{B}$.

x in Q means that it is in $(A \cap B_i)$ for some B . This means that x is in A . x is also in B_i , so x is in $\cup \mathcal{B}$

All done!

3 Functions

Recall we refer to all the combinations of two sets X and Y with $X \times Y$.

A function f that maps X into Y $f : X \rightarrow Y$ is a relation $f \subseteq X \times Y$ such that

- for every $x \in X$, there exists a $y \in Y$ such that $f(x) = y$.
- For every $y, z \in Y$ with $f(x) = y$ and $f(x) = z$, we have $y = z$

X is called the domain of f ; Y is the codomain of f ; the range of f is

$$f(X) := \{y \in Y : f(x) = y \text{ for some } x \in X\}$$

General Rule: If X and Y are a nonempty sets, $f : X \rightarrow Y$, then for any $\mathcal{A} \subseteq 2^X$

$$f(\cup \mathcal{A}) = \cup \{f(A) : A \in \mathcal{A}\}$$

E.g.

$$f = x^2$$

$$X = \{1, 2, 3, 5\}$$

$$\mathcal{A} = \{\{1, 2\}, \{1, 3\}\}$$

$$\cup \mathcal{A} = \{1, 2, 3\}$$

$$f(\cup \mathcal{A}) = \{1, 4, 9\}$$

$$\cup \{f(A) : A \in \mathcal{A}\} = \{1, 4\} \cup \{9\}$$

However: If X and Y are a nonempty sets, $f : X \rightarrow Y$, then for any $\mathcal{A} \subseteq 2^X$

$$f(\cap \mathcal{A}) \subseteq \cap \{f(A) : A \in \mathcal{A}\}$$

E.g. $f = x^2$

$$X = \{-2, 2\}$$

$$\mathcal{A} = \{\{2\}, \{-2\}\}$$

$$\cap \mathcal{A} = \emptyset$$

$$f(\cap \mathcal{A}) = \emptyset$$

$$\cap \{f(A) : A \in \mathcal{A}\} = \{4\} \cap \{4\} = \{4\}$$